Name: Test Case - fully correct Class: 10-B Subject: Combined Paper Total: 15 marks

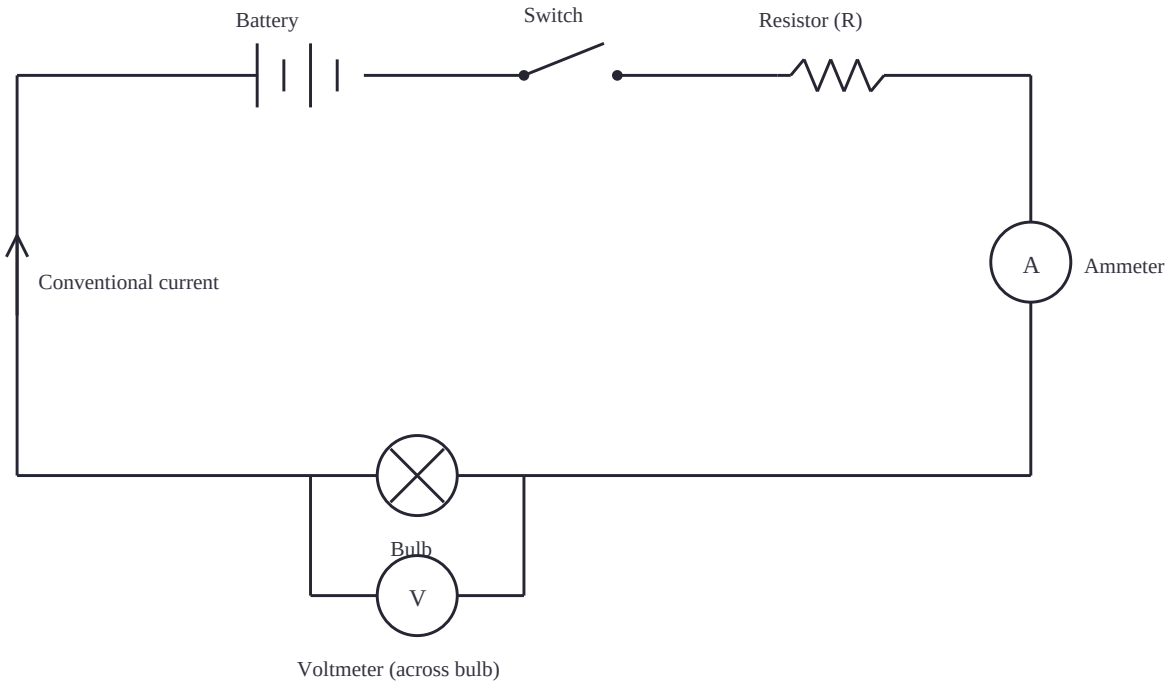
Question 1

A simple electric circuit provides a closed path through which electric current can flow. The battery provides the potential difference that drives the current around the circuit. The switch opens or closes the circuit: when it is closed there is a complete conducting path and current flows through the bulb and the resistor, and when it is open the path is broken and no current flows.

The battery, switch, resistor, bulb and ammeter are all connected in series in the main circuit. The ammeter is connected in series because it measures the current flowing through the circuit. The voltmeter is connected in parallel across the bulb, because it measures the potential difference between the two ends of the bulb.

The amount of current depends on the potential difference and on the resistance. By Ohm's law, V = IR. If the voltage of the battery stays constant and the resistance is increased, the current flowing through the circuit decreases; reducing the resistance allows more current to flow.

The diagram below is labelled with every component and shows the direction of conventional current from the positive terminal of the battery through the external circuit back to the negative terminal.



Question 2

In my opinion technology does not automatically make students better or worse learners; its effect depends entirely on how it is used. Information that was once difficult to reach is now available in seconds, and a student who cannot follow a topic in class can watch several explanations until one of them makes sense. For example, a student struggling with the water cycle can compare three short videos and find the explanation that finally fits.

However, easy access also creates a real danger. Students may become dependent on searching for an answer instead of working a problem out for themselves, and a student who looks up every difficult question may finish the homework without ever developing the reasoning that the homework was set to build. There is also the risk of accepting inaccurate information simply because it appears online.

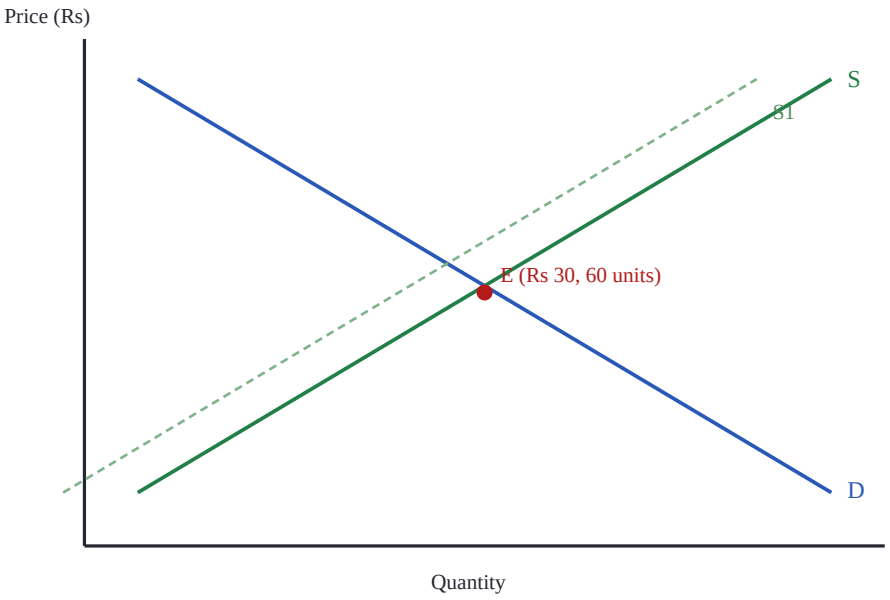
In conclusion, technology is most useful when it is used to understand concepts, explore ideas and check one's

own work, rather than to copy answers. It should therefore be treated as a tool that supports thinking, not as a replacement for thinking.

Question 3

I have plotted the schedule with the quantity on the horizontal axis and the price on the vertical axis. The demand curve slopes downward from left to right because consumers demand a greater quantity at a lower price, and the supply curve slopes upward from left to right because producers are willing to supply more at a higher price.

The two curves intersect at a price of Rs 30 and a quantity of 60 units. This is the market equilibrium, because at this price the quantity demanded is exactly equal to the quantity supplied.

If the market price is below the equilibrium price, the quantity demanded is greater than the quantity supplied, which creates a shortage and puts upward pressure on the price. If the market price is above the equilibrium price, the quantity supplied is greater than the quantity demanded, which creates a surplus and puts downward pressure on the price.

If the cost of production increases, producing becomes less profitable at every price, so producers supply less at each price and the supply curve shifts to the left. The new equilibrium is therefore at a higher price and a lower quantity, assuming demand is unchanged.